

PMM3 vs MLE-TN: Comparison of Semiparametric and Parametric Estimation of Linear Regression Parameters under Two-piece Normal Distributed Errors

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Abstract. This paper addresses the problem of estimating the parameter vector of a linear regression model under symmetric non-Gaussian errors following a Two-piece Normal (TN) distribution. Three methods are compared: Ordinary Least Squares (OLS), Maximum Likelihood Estimation for the TN distribution (MLE-TN), and the third-power Polynomial Maximization Method (PMM3). MLE-TN is fully parametric; PMM3 is semiparametric, using only the 2nd, 4th and 6th-order moments. Theoretical analysis shows that PMM3 achieves up to a 95.5% variance reduction compared to OLS at $\lambda = 5$ (the TN shape parameter), while MLE-TN provides up to a 28-fold advantage for large λ . A Monte Carlo study confirmed that the empirical efficiency coefficient g_3 converges to the theoretical value for $n \geq 100$. PMM3 is shown to be robust under misspecification, whereas MLE-TN can be inefficient. On real-world data (*Iris versicolor*), PMM3 achieves a 43% bootstrap variance reduction and improves LOO-CV by 2.3% over OLS. An empirical efficiency threshold for PMM3 is established at $\gamma_4 \lesssim -0.7$.

Keywords: Polynomial Maximization Method, Linear Regression, Two-piece Normal distribution, Maximum Likelihood Estimation, Estimation Efficiency, Robustness, Monte Carlo.

1 Introduction

The Ordinary Least Squares (OLS) method [5] is the Best Linear Unbiased Estimator under Gaussian errors, but its efficiency depends on the shape of the error distribution [2]. When normality is violated, OLS loses its asymptotic optimality, whereas estimators exploiting higher-order moments may achieve lower variance.

In practical applications, including metrology, medicine, econometrics, and the natural sciences, measurement errors frequently follow symmetric but non-Gaussian distributions. Platykurtic distributions ($\gamma_4 < 0$) commonly arise in measurements with

a limited operating range, in regression residual histograms, and when multiple sources of uniformly distributed errors are aggregated.

One important family of such distributions is the Two-piece Normal (TN) distribution, introduced in [8] and further investigated in [14]. The TN distribution covers a broad spectrum of symmetric shapes, ranging from an approximately Gaussian form at ($\lambda \approx 0$) to a strongly bimodal form at ($\lambda \gg 0.707$), making it a natural benchmark for studying symmetric platykurtic distributions with $\gamma_4 \in (-2, 0)$.

Three fundamentally different approaches can be distinguished: Parametric approach (MLE-TN): jointly estimates all parameters (β, η, λ) , attains the Cramér–Rao lower bound under correct model specification, but is sensitive to model misspecification [6], [18]; Semiparametric approach (PMM3): utilizes only the 2-nd, 4-th and 6-th order moments. It does not require knowledge of the functional form of the error distribution and remains robust under model misspecification; Nonparametric and robust approaches: including Weighted Least Squares (WLS), Least Absolute Deviations (LAD) regression [13], quantile regression [9], and adaptive Maximum Likelihood Estimation [15].

The present study addresses the following practical question: under what conditions, and to what extent, does PMM3 outperform both MLE-TN and OLS.

1.1 Literature Review

In [2], a general expression for the asymptotic relative efficiency of OLS was derived in terms of the cumulants of the underlying distribution. Subsequently, [16] systematically investigated parameter estimation in linear regression models with symmetric non-Gaussian errors. The studies reported in [1] and [4] confirmed that, for $\gamma_4 < -0.5$, classical OLS may perform worse than the optimal estimator. Robust regression techniques [6] and LAD regression [13] are primarily designed for leptokurtic ($\gamma_4 > 0$) distributions. In contrast, for platykurtic ($\gamma_4 < 0$) distributions, none of the standard methods can be regarded as optimal. In [12], a semiparametric regression framework with an unspecified error distribution was developed and shown to outperform MLE when the assumption of normality is misspecified. In [14], both Bayesian and frequentist approaches to linear regression with TN-distributed errors were proposed, and MLE estimation was investigated using L-BFGS-B optimization. Furthermore, [11] developed an objective Bayesian methodology for the broader class of two-piece distributions. Both studies emphasize that, under model misspecification, parametric approaches may be inferior to semiparametric alternatives [18].

As for the Polynomial Maximization Method (PMM), it was developed by Kunchenko and Lega [10]. The theoretical efficiency coefficient of PMM3 derived in [17] is given in Section 2.4.

In [20], PMM was extended to linear regression with asymmetric errors (PMM2, $g_2 = 1 - \gamma_3^2 / (2 + \gamma_4)$) and in [19], the first application of the PMM3 to linear regression with symmetric non-Gaussian errors of general form was presented. In [7], a normality test based on the sample skewness and kurtosis was proposed. The corresponding

adaptive decision algorithm is: normally distributed residuals – apply OLS; asymmetric residuals – apply PMM2; symmetric platykurtic residuals – apply PMM3.

1.2 Scientific contribution of the study

Although previous studies have investigated PMM3 for general symmetric distributions and MLE-TN for the Two-piece Normal model, a systematic direct comparison between PMM3 and MLE-TN within the TN framework has not previously been conducted. The principal contributions of this work are as follows:

(i) Theoretical comparison: derivation and analysis of analytical expressions for $g_3(\lambda)$, $ARE(\text{MLE-TN})(\lambda)$, and the efficiency curve comparing PMM3 and MLE-TN; (ii) Monte Carlo validation: $\lambda \in \{0.5, 1, 1.5, 2, 3, 5\}$, $n \in \{46, 100, 200, 500\}$, using $M = 500$ replications; (iii) Robustness analysis: demonstration that PMM3 maintains $ARE \approx 1$ under model misspecification, whereas MLE-TN may experience a substantial loss of efficiency; (iv) Numerical stability: verification that the Newton-Raphson algorithm converges in 100% of the replications across all 24 Monte Carlo scenarios; (v) Empirical threshold $\gamma_4 \lesssim -0.7$: establishment and justification of an empirical efficiency threshold based on seven real-world datasets; (vi) Real-data application: analysis of the Iris versicolor dataset ($n = 50$, $\gamma_4 = -0.966$), demonstrating that PMM3 achieves a 43% reduction in variance.

2 Materials and methods

2.1 Linear regression model

Consider the following linear regression model:

$$y_v = x_v^T \beta + \varepsilon_v, \quad v = 1, \dots, n, \quad (1)$$

where y_i denotes the response variable, $x_i \in \mathbb{R}^p$ is the vector of regressors, $\beta \in \mathbb{R}^p$ is the vector of unknown regression parameters, and ε_i are independent and identically distributed random errors.

Assumption. For PMM3: the error term satisfies $\varepsilon_v \sim$ a symmetric distribution with $E(\varepsilon) = 0$ and possesses finite moments up to the sixth order. **For MLE-TN:** the error term follows the Two-piece Normal distribution, $\varepsilon_v \sim TN(0, \eta, \lambda)$.

2.2 Two-piece Normal (TN) distribution

The Two-piece Normal (TN) distribution with zero location parameter, scale parameter $\eta > 0$, and shape parameter $\lambda > 0$ has the probability density function [14]:

$$f(z | \lambda) = \frac{1}{\sqrt{2\pi}} \cdot \exp\left(-\frac{z^2}{2} - \frac{\lambda^2}{2}\right) \cdot \cosh(\lambda z), \quad z \in \mathbb{R}. \quad (1)$$

The principal properties of the TN distribution are as follows: $\lambda \rightarrow 0$ – the distribution approaches the standard normal distribution $N(0, 1)$; $\lambda > 1/\sqrt{2}$ – the distribution becomes bimodal, exhibiting two symmetric modes; γ_4 is strictly negative for all values of λ , indicating a platykurtic distribution.

The normalized excess kurtosis is given by:

$$\gamma_4 = \frac{\mu_4}{\mu_2^2} - 3 < 0 \quad \text{for all } \lambda > 0. \quad (1)$$

2.3 MLE-TN: Maximum likelihood estimation

The MLE-TN approach jointly estimates the parameter vector (β, η, λ) by maximizing the log-likelihood function:

$$l(\beta, \eta, \lambda). \quad (1)$$

The optimization procedure is performed using the L-BFGS-B algorithm with the following initial values: $\beta_0 = \hat{\beta}_{OLS}$, $\eta_0 = \exp(\log SD(\hat{e}_{OLS}))$, $\lambda_0 = 1.0$. Advantages: achieves the Cramér–Rao lower bound under correct model specification; provides $ARE(MLE-TN/OLS) \gg 1$ for sufficiently large values of λ . Limitations: requires estimation of $p+2$ parameters; is sensitive to model misspecification; may exhibit numerical instability for small sample sizes and large n values of λ .

2.4 PMM3: Polynomial Maximization Method (S = 3)

The PMM3 estimator evaluates only the regression parameter vector β by employing a moment-corrected score function. The algorithm consists of the following steps: compute the initial estimate $\beta^{(0)} = \hat{\beta}_{OLS}$ and evaluate the sample central moments of the residuals m_2, m_4, m_6 , then the correction coefficient is estimated once according to

$$\kappa = \frac{m_6 - 3 m_4 m_2}{m_4 - 3 m_2^2}. \quad (1)$$

If the corresponding condition is violated $|m_4 - 3 m_2^2| < 10^{-10}$, the algorithm returns the Gaussian fallback solution $\hat{\beta}_{OLS}$.

The parameter vector is updated according to the Newton-Raphson procedure: $Z_{1v} = \varepsilon_v(\kappa - \varepsilon_v^2)$; $Z = X^T Z_1$; $J = X^T \text{diag}(3\varepsilon^2 - \kappa) X$; $\beta \leftarrow \beta - J^{-1} Z$. The iterations continue until the prescribed convergence criterion is satisfied: $\|Z\|_2 < 10^{-6}$. In practice, convergence is typically achieved within 3-5 iterations. To ensure numerical stability, if the specified divergence condition is met, $\|\beta - \hat{\beta}_{OLS}\| > 10$ the algorithm returns the initial estimate $\hat{\beta}_{OLS}$.

The theoretical efficiency coefficient of PMM3 is

$$g_3 = 1 - \frac{\gamma_4^2}{6 + 9\gamma_4 + \gamma_6}, \quad (1)$$

where $\gamma_4 = \mu_4/\mu_2^2 - 3$, $\gamma_6 = \mu_6/\mu_2^3 - 15(\mu_4/\mu_2^2) + 30$ – normalized fourth and sixth-order cumulants. An important advantage of PMM3 is that it does not require knowledge of the distribution: cumulants are estimated directly from the sample.

2.5 Asymptotic normality of PMM3

Under the standard regularity conditions, the PMM3 estimator is consistent and asymptotically normally distributed. $\sqrt{n}(\hat{\beta}_{PMM3} - \beta) \rightarrow^d N(0, g_3 \cdot \sigma^2 \cdot (X^T X / n)^{-1})$ Its asymptotic covariance matrix is reduced by a factor of $(1/g_3)$ relative to that of the OLS estimator: $\text{Var}(\hat{\beta}_{PMM3}) = g_3 \cdot \text{Var}(\hat{\beta}_{OLS})$

For the Gaussian case ($\gamma_4 = 0$) $g_3 = 1$ PMM3 is asymptotically equivalent to OLS. For platykurtic distributions ($\gamma_4 < 0$) $g_3 < 1$ PMM3 is more efficient than OLS. The consistency of PMM3 depends only on the assumption of zero-mean errors and does not require specification of the exact distributional form. A more detailed derivation is provided in [10] and [20]. The three methods differ in what they must assume. OLS and PMM3 both estimate only the p coefficients of the vector and require no knowledge of the distributional form, whereas MLE-TN estimates $p + 2$ parameters and requires the TN form to be correctly specified. Under TN errors OLS is the baseline ($ARE = 1$), PMM3 attains $g_3 < 1$ whenever the fourth-order cumulant is negative, and MLE-TN attains the Cramér–Rao bound.

2.6 Theoretical characteristics of the TN distribution

The shapes of the TN density are shown in the left panel of Fig. 1 and the theoretical efficiency curve $g_3(\lambda)$ in the right panel. Evaluated at the six shape values used throughout this study, the moment formulas give $\gamma_4 = -0.080, -0.500, -0.959, -1.280, -1.620$ and -1.849 with $g_3 = 0.9988, 0.9286, 0.6574, 0.3868, 0.1490$ and 0.0454 for $\lambda = 0.5, 1.0, 1.5, 2.0, 3.0$ and 5.0 , corresponding to a variance reduction relative to OLS of

0.1%, 7.1%, 34.3%, 61.3%, 85.1% and 95.5%. The distribution is bimodal for $\lambda > 1/\sqrt{2}$, that is for every $\lambda \geq 1.5$ in this set.

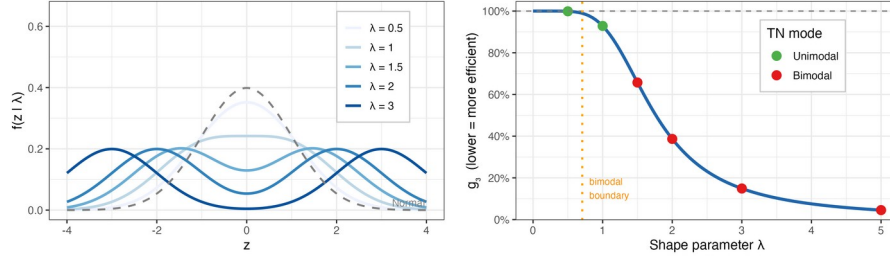


Fig. 1. Two-piece Normal model. Left: TN densities for $\lambda \in \{0.5, 1, 1.5, 2, 3\}$, standard normal dashed. Right: theoretical efficiency g_3 as a function of λ ; the dotted line marks the bimodality boundary $\lambda = 1/\sqrt{2}$, unimodal cases green, bimodal red.

3 Results and discussion

3.1 Monte Carlo study

The data-generating model is defined as:

$$y_v = 1 + \beta_1 x_v + \varepsilon_v, \quad \varepsilon_v \sim TN(0, 1, \lambda). \quad (1)$$

The simulation parameters: number of replications $M=500$; sample sizes $n \in \{46, 100, 200, 500\}$; shape parameters $\lambda \in \{0.5, 1, 1.5, 2, 3, 5\}$; random seed = 20260312. Metrics: Bias= $\hat{\beta}_1 - 1$; Var($\hat{\beta}_1$); MSE=Bias²+Var; ARE=Var(OLS)/Var(method); $g_{3_emp} = \text{Var(PMM3)}/\text{Var(OLS)}$.

The results for all 24 scenarios are summarised in Figs. 2 and 3; the full numeric table is provided as supplementary material.

Here ARE > 1 indicates superiority over OLS. Furthermore $g_{3_emp} = \text{Var(PMM3)}/\text{Var(OLS)}$, $\Delta g_3 = (g_{3_emp} - g_{3_theor})/g_{3_theor} \times 100\%$. For $n \geq 100$ the deviation remains within $< \pm 10\%$.

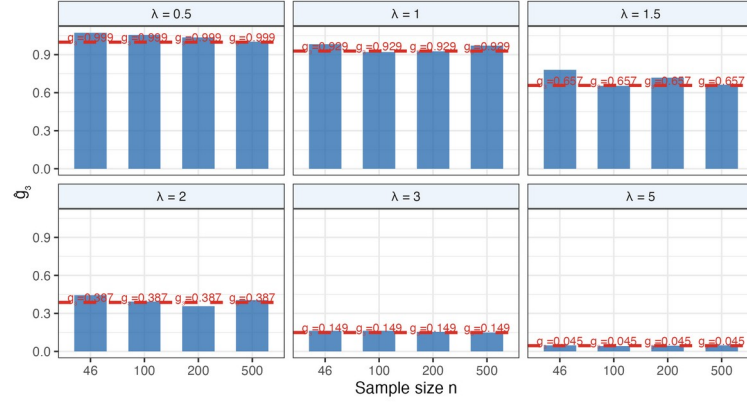


Fig. 2. Empirical efficiency coefficient $\hat{g}_3$ (bars) against its theoretical value (dashed line) for all 24 Monte Carlo scenarios (six values of $\lambda \times$ four sample sizes n , $M = 500$).

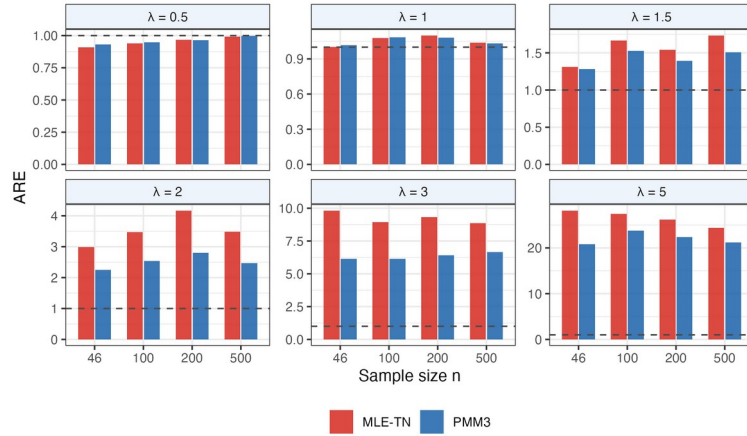


Fig. 3. Monte Carlo ARE of MLE-TN and PMM3 relative to OLS, for six shape values λ and four sample sizes n ($M = 500$). ARE > 1 means more efficient than OLS.

PMM3 significantly outperforms OLS at $\lambda \geq 1.5$: ARE ranging from 1.5 to 23.8 – the theory has been confirmed. MLE-TN is even more effective than PMM3 for $\lambda \geq 1.5$ (assuming the TN specification is correct), but it estimates $p + 2$ parameters – excessive parameterization for small n . The empirical efficiency coefficient $\hat{g}_3_{emp}$ approximates its theoretical value $\hat{g}_3_{theor}$ with an accuracy of approximately $\pm 10\%$ for sufficient sample sizes ($n \geq 100$). For the most challenging scenario, $\lambda = 5$, $n = 100$, the efficiencies were ARE(PMM3) = 23.8 compared with ARE(MLE-TN) = 27.4. Thus, PMM3 lags behind MLE-TN by only about 15%, while requiring no prior knowledge of the underlying error distribution.

3.2 Real-world data: Fisher's iris dataset (Iris versicolor)

A subset of the classical Fisher Iris dataset [3] corresponding to the species Iris versicolor was used for empirical evaluation. The dataset contains $n = 50$ observations.

$$Sepal.Length_v = \beta_0 + \beta_1 \cdot Sepal.Width_v + \varepsilon_v. \quad (1)$$

OLS Residual Diagnostics: $n=50$; $\hat{\gamma}_3=0.273$ (indicating only negligible asymmetry); $\hat{\gamma}_4 = -0.966$ (indicates pronounced platykurticity); $span / SD \approx 3.58$ – suggesting a restricted measurement range.

The corresponding theoretical efficiency is: $g_{3theor} = 0.517$ which corresponds to a 51.7% reduction in variance. Based on the residual diagnostics: $\gamma_4 = -0.966 < -0.7$ PMM3 is recommended.

Table 1. Comparison of estimation methods for iris versicolor ($n = 50$).

Method	$\hat{\beta}_1$	LOO-MSE	g_3_emp (bootstrap)
OLS	0.865	0.2070	1.000 (baseline)
PMM3	0.797	0.2022	0.570
MLE-TN	0.753	0.2109	0.746

The relative variance is computed as $g_3_emp = \text{Var}(\text{method})/\text{Var}(\text{OLS})$ using bootstrap estimation with $B = 2000$ replications.

PMM3 achieved the smallest prediction error: LOO-MSE = 0.2022, which is lower than that obtained by OLS = 0.2070 and MLE-TN = 0.2109. Bootstrap analysis indicates a 43% reduction in variance ($g_3_emp = 0.570$), compared with the theoretical prediction of 51.7%. Given the relatively small sample size ($n = 50$), this agreement can be considered satisfactory. Despite the additional parameters (η, λ) , MLE-TN exhibited the poorest performance. Consequently, for datasets characterized by $n = 50$ and $\gamma_4 \approx -0.97$, PMM3 represents the optimal estimation strategy.

Negative example. For the whiteside dataset (MASS package, household gas consumption versus outdoor temperature, $n = 56$) the residuals give $\gamma_4 = -0.431$, i.e. above the threshold: the theoretical gain is only 8.0% ($g_3\ theor = 0.920$), the bootstrap ratio is $g_3\ emp = 0.978$ and LOO-MSE PMM3/OLS = 0.999. Together with four further real datasets with $\gamma_4 \in [-0.54, -0.12]$, for which $g_3\ emp \in [0.97, 1.03]$, this supports the empirical threshold $\gamma_4 \lesssim -0.7$.

3.3 Robustness under model misspecification

Robustness was assessed by generating data from non-TN error distributions (Table 2), with sample size $n = 100$, number of Monte Carlo replications $M = 500$.

Table 1. ARE under model misspecification ($n = 100, M = 500$).

True DGP	γ_4 (true)	ARE(MLE-TN)	ARE(PMM3)
Uniform	-1.200	1.655	1.000
Triangular	-0.600	0.969	1.000
Logistic	+1.200	0.994	1.000
Student's t10	+1.000	0.982	1.000

PMM3 is robust, maintaining: $ARE \approx 1.0$ for all investigated distributions. Consequently, even when its assumptions are not fully satisfied, PMM3 performs no worse than OLS. MLE-TN is considerably more sensitive to model misspecification. For example, under the triangular distribution, $ARE = 0.969$ indicating performance that is actually inferior to OLS. Therefore, PMM3 should be regarded as the safer estimation method whenever the true error distribution is uncertain. The Newton–Raphson implementation converged in 100% of the 500 replications with no NA outcomes for every $\lambda \in \{0.8, 1.0, 1.2, 1.5, 2.0, 3.0\}$, including the strongly bimodal cases, with absolute bias never exceeding 0.022.

4 Conclusions

1. Theoretical results. PMM3 provides a 7% to 95% reduction in variance relative to OLS for TN-distributed errors with $\lambda \in \{1, \dots, 5\}$. Under correct model specification, MLE-TN can achieve an even greater efficiency gain; however, it requires estimation of $p+2$ parameters. Monte Carlo study. MLE-TN is substantially more efficient than PMM3 for large values of λ , provided that the TN specification is correct. The empirical efficiency coefficient g_3 of PMM3 converges to its theoretical value for sample sizes of approximately $n \geq 100 - 200$.
2. Robustness. PMM3 consistently maintains $ARE \approx 1$ under model misspecification, whereas MLE-TN may perform worse than OLS. PMM3 is therefore the safe choice when the error distribution is uncertain.
3. Real-world data. For the Iris versicolor dataset ($\gamma_4 = -0.966, n = 50$) PMM3 achieves a 43% reduction in variance, while its predictive performance satisfies $LOO-MSE \text{ PMM3} < OLS < MLE-TN$.
4. Empirical threshold. PMM3 demonstrates a practical advantage over OLS only when $\gamma_4 \lesssim -0.7$. For $-0.7 < \gamma_4 < 0$ all three methods are, for practical purposes, equivalent to OLS. In practice: prefer MLE-TN when the TN model is justified and n is large, PMM3 when the error distribution is uncertain, and OLS for approximately Gaussian errors.

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